# DG-HMCF vs Normal Attention Mechanism — Detailed Comparison
# Source: PHD WORK/DG-HMCF/models/

| Component | Normal Attention | DG-HMCF (This Work) | Source File |
|---|---|---|---|
| Fusion Type | Single-level self-attention | Hierarchical 6-pair cross-modal | dg_hmcf.py:Line 97 |
| Gating Mechanism | Fixed equal weights | Dynamic Reliability Gating (learned per-sample) | dynamic_gating.py |
| Temporal Modeling | Single scale only | Multi-Scale (kernels 3,5,7) per modality | multiscale_temporal.py |
| Cross-Modal Interaction | Single-pass self-attention | Bidirectional pairwise (6 pairs x 2 directions) | hierarchical_cross_modal.py |
| Missing Modality | Fails or ignores | Learned imputation + modality dropout augmentation | missing_modality.py |
| Output | Single class logit | Binary label + PHQ-8 score (multi-task) | classifier.py |
| Explainability | None | Modality importance + attention map visualization | explainability.py |
| Temperature Scaling | None (fixed softmax) | Learnable temperature tau (req_grad=True) | dynamic_gating.py:Line 36 |
| Context Aggregation | None | Global context modulates per-modality quality scores | dynamic_gating.py:Line 64 |
| Parameter Count | Shared projection | Independent MLPs per modality + cross-attn per pair | dg_hmcf.py:count_parameters() |

*DG-HMCF is NOT a normal attention mechanism. It uses Dynamic Reliability Gating, Hierarchical 6-pair Cross-Modal Attention, Multi-Scale Temporal Fusion, Missing Modality Imputation, and Multi-Task output (binary + PHQ-8 regression). Each component is independently implemented in PHD WORK/DG-HMCF/models/modules/*